

A Lean-Verified 51/50 Cutoff for Four-Arc Minimizers

Proof built, Lean-verified, and semantically accepted by
Agentic Lean Math Assistant · github.com/RosaRojacr/agentic-lean-math-assistant

1. Problem and result

Cañete, Miranda Jr., and Vittone study the isoperimetric problem for the planar density that equals 1 on the strip $\{|y| \leq 1\}$ and a constant $\lambda > 1$ outside it [1, §3.2]. Their Lemma 3.8 reduces the problem to four geometric candidate families. Numerical evidence following equations (26)–(27) suggests that the symmetric type-(iv) four-arc family is never minimizing, but Theorem 3.16 proves its exclusion only in a high-density range. The result below gives a Lean-verified exclusion for the explicit four-arc model whenever $\lambda \geq 51/50$.

Fix $\lambda > 1$ and define

$$f_\lambda(x, y) = \begin{cases} 1, & |y| \leq 1, \\ \lambda, & |y| > 1. \end{cases}$$

For a measurable carrier $E \subset \mathbb{R}^2$, set

$$A_\lambda(E) = \int_E f_\lambda \, d\mathcal{L}^2, \quad P_\lambda(E) = \int_{\partial E} f_\lambda \, d\mathcal{H}^1.$$

Here ∂E is the complete topological frontier, and $\mathcal{H}^1$ is Euclidean one-dimensional Hausdorff measure. The interfaces $|y| = 1$ have density 1. These choices are part of the formal statement: in particular, no interior chord is counted as exposed boundary, and Hausdorff measure is taken only after the coordinate plane is identified with its Euclidean realization.

LEAN REFERENCES

`CMVGeometry.lean` StripDensity; WeightedArea; FrontierMeasure; WeightedPerimeter; planeEuclideanHomeomorph

The comparison class is not restricted to four-arc curves. It contains general measurable carriers with integrable weighted area and complete-frontier weighted perimeter, together with the explicit type-(iii) and cap-replacement competitors used below. The package’s term “finite perimeter” refers to these recorded integrability conditions; it does not identify the model with the full theory of Caccioppoli sets modulo null sets.

LEAN REFERENCES

`CMVGeometry.lean` FinitePerimeterRegion; AdmissibleCompetitor; AdmissibleCompetitor.weightedArea_eq_carrier; AdmissibleCompetitor.WeightedPerimeter

Theorem. Let C be a modeled type-(iv) four-arc carrier with curvature h and exterior half-angle α , where

$$0 < h \leq 1, \quad 0 < \alpha < \frac{\pi}{2}, \quad \lambda \cos \alpha = h.$$

If $\lambda \geq 51/50$, then C is not a weighted-perimeter minimizer. More precisely, there exists an admissible carrier with the same weighted area as C and strictly smaller weighted perimeter.

LEAN REFERENCES

```
FourArcCandidate.lean FourArcCandidate.SatisfiesCMVTypeIVHypotheses;
                        FourArcCandidate.IsWeightedPerimeterMinimizer
CMVPublishedCutoff51_50.lean CMVPublishedCutoff51_50.candidate_not_isWeightedPerimeterMinimizer_from_51_50
```

The endpoint $h = 1$ is included. The hypotheses above encode the density jump, the principal-angle range, and the Snell incidence relation; they do not assume minimality, a numerical certificate, or a perimeter comparison. Thus the theorem addresses the explicit model extracted from [1], rather than importing the source paper’s unformalized candidate-classification argument.

2. Geometry and scalar reduction

The carrier consists of a central strip core and congruent exterior circular caps above and below the strip. Put

$$R = \frac{1}{h}, \quad c = \frac{2\sin\alpha}{h}.$$

Each exterior cap has chord length c and area $(\alpha - \sin\alpha\cos\alpha)/h^2$. The two strip arcs have half-angle $\arcsin h$, while the incidence relation gives $\alpha = \arccos(h/\lambda)$.

LEAN REFERENCES

```
FourArcCandidate.lean FourArcCandidate.capChord; FourArcCandidate.capArea; FourArcCandidate.stripCore;
                        FourArcCandidate.assembly; FourArcCandidate.alpha_eq_arccos
```

These formulas are proved from the coordinate carriers. The formal development integrates the circular segments, identifies the complete frontier, computes the Euclidean Hausdorff length of each arc, and proves that the finitely many junction points have zero $\mathcal{H}^1$ measure. It therefore derives, rather than stipulates, the weighted area and perimeter formulas used in the comparison.

LEAN REFERENCES

```
CMVGeometry.lean FourArcAssembly.weightedArea_formula; FourArcAssembly.frontier_carrier
FrontierPerimeter.lean fourArc_frontier_weightedPerimeter_eq
FourArcCandidate.lean FourArcCandidate.candidate_area_formula; FourArcCandidate.candidate_perimeter_formula
```

For fixed $\lambda > 1$, write $q = 2s - 1$ for $0 < s < 1$, and define the type-(iii) quantities

$$B_3(s) = \lambda \arccos\left(\frac{q}{\lambda}\right) + \arcsin q + \frac{\pi}{2},$$

$$D_3(s) = \frac{\sqrt{\lambda^2 - q^2}}{\lambda} - \sqrt{1 - q^2}.$$

For $0 < t \leq 1$, define the type-(iv) quantities

$$B_4(t) = \lambda \arccos\left(\frac{t}{\lambda}\right) + \arcsin t,$$

$$D_4(t) = \frac{\sqrt{\lambda^2 - t^2}}{\lambda} - \sqrt{1 - t^2}.$$

The corresponding areas and perimeters are

$$A_3(s) = \frac{B_3(s) + (2s+1)D_3(s)}{s^2}, \quad P_3(s) = \frac{2(B_3(s) + D_3(s))}{s},$$

$$A_4(t) = \frac{2(B_4(t) + tD_4(t))}{t^2}, \quad P_4(t) = \frac{4B_4(t)}{t}.$$

LEAN REFERENCES

```
LeanSuffixAnalytic.lean LeanSuffixAnalytic.typeThreeShape; LeanSuffixAnalytic.typeThreeAngle;
LeanSuffixAnalytic.typeThreeDelta; LeanSuffixAnalytic.typeThreeArea;
LeanSuffixAnalytic.typeThreePerimeter; LeanSuffixAnalytic.typeFourAngle;
LeanSuffixAnalytic.typeFourDelta; LeanSuffixAnalytic.typeFourArea;
LeanSuffixAnalytic.typeFourPerimeter
```

For every $s \in (0, 1)$, the type-(iii) construction realizes $A_3(s)$ and $P_3(s)$ by an admissible carrier: one exterior cap, two strip arcs, and the exposed lower segment. The construction includes the major-cap branch for $s < 1/2$ and the degenerate lower segment at $s = 1/2$. Both are essential because an equal-area root need not remain above $1/2$.

LEAN REFERENCES

```
TypeThreeAssembly.lean TypeThreeAssembly; TypeThreeAssembly.frontier_carrier;
TypeThreeAssembly.weightedArea_formula; TypeThreeAssembly.weightedPerimeter_formula;
TypeThreeAssembly.hasModeledCompetitor
CMVSuffixModel.lean CMVSuffixModel.TypeThreeAssembly.scalarWeightedArea_eq_typeThreeArea;
CMVSuffixModel.TypeThreeAssembly.scalarWeightedPerimeter_eq_typeThreePerimeter
```

The modeled four-arc carrier satisfies

$$A_\lambda(C) = A_4(h), \quad P_\lambda(C) = P_4(h).$$

It is therefore enough to find $s \in (0, 1)$ such that $A_3(s) = A_4(h)$ and $P_3(s) < P_4(h)$. The realization theorem then converts this scalar inequality into an admissible geometric competitor.

LEAN REFERENCES

```
CMVSuffixModel.lean CMVSuffixModel.FourArcCandidate.weightedArea_eq_typeFourArea;
CMVSuffixModel.FourArcCandidate.weightedPerimeter_eq_typeFourPerimeter;
CMVSuffixModel.FourArcCandidate.not_isWeightedPerimeterMinimizer_of_analytic_typeThree
```

3. The stationary comparison

Define the fold numerators

$$K_i(u) = u^3 A_i(u), \quad 0 < u < 1.$$

With $q = 2s - 1$, direct differentiation gives

$$K_3(s) = 4s \left(\frac{1+q}{\sqrt{1-q^2}} - \frac{\lambda^2+q}{\lambda\sqrt{\lambda^2-q^2}} \right) - 2(B_3(s) + D_3(s)),$$

$$K_4(t) = 4 \left[t \left(\frac{1}{\sqrt{1-t^2}} - \frac{\lambda}{\sqrt{\lambda^2-t^2}} \right) - B_4(t) \right].$$

Moreover,

$$P_i(u) = u A_i(u) = \frac{K_i(u)}{u^2}.$$

LEAN REFERENCES

```
LeanSuffixAnalytic.lean LeanSuffixAnalytic.typeThreeFold; LeanSuffixAnalytic.typeFourFold;
LeanSuffixAnalytic.hasDerivAt_typeThreeArea; LeanSuffixAnalytic.hasDerivAt_typeFourArea;
LeanSuffixAnalytic.hasDerivAt_typeThreePerimeter;
LeanSuffixAnalytic.hasDerivAt_typeFourPerimeter
```

For $|z| < 1$,

$$\frac{1}{(1 - z^2)^{3/2}} - \frac{\lambda}{(\lambda^2 - z^2)^{3/2}} > 0.$$

Indeed, $\lambda^2 - z^2 > \lambda^2(1 - z^2)$ away from $z = 0$, and at $z = 0$ the difference is $1 - \lambda^{-2} > 0$. Consequently both folds are strictly increasing. The unique zero t_* of K_4 is the minimum point of A_4 on $(0, 1]$, including the one-sided endpoint $t = 1$.

LEAN REFERENCES

```
LeanSuffixAnalytic.lean LeanSuffixAnalytic.hasDerivAt_typeThreeFold; LeanSuffixAnalytic.hasDerivAt_typeFourFold;
LeanSuffixAnalytic.typeThreeFold_strictMonoOn;
LeanSuffixAnalytic.typeFourFold_strictMonoOn; LeanSuffixAnalytic.typeFourArea_fold_le
```

A stationary certificate supplies $s_*, t_* \in (0, 1)$ satisfying

$$\frac{1}{2} < s_* < 1, \quad A_3(s_*) = A_4(t_*), \quad K_4(t_*) = 0,$$

and separately proves

$$K_3(s_*) < 0, \quad P_3(s_*) < P_4(t_*).$$

LEAN REFERENCES

```
LeanSuffixAnalytic.lean LeanSuffixAnalytic.StationaryEqualAreaPair
```

Because K_3 is increasing and negative at s_* , the function A_3 is strictly decreasing on $(0, s_*]$; moreover, $A_3(s) \rightarrow +\infty$ as $s \downarrow 0$. Hence every $t \in (0, 1]$ determines a unique descending-branch root $r(t) \in (0, s_*]$ with

$$A_3(r(t)) = A_4(t).$$

LEAN REFERENCES

```
LeanSuffixAnalytic.lean LeanSuffixAnalytic.typeThreeArea_tendsto_atRight_zero;
LeanSuffixAnalytic.typeThreeArea_descending_root_existsUnique;
LeanSuffixAnalytic.typeThreeArea_descending_root_existsUnique_of_stationary_pair
EqualAreaEnvelope.lean LeanSuffixAnalytic.DescendingEqualAreaEnvelope
```

The exact same-curvature inequality $A_3(u) < A_4(u)$, valid for $0 < u \leq 1$, then yields the strict root order $r(t) < t$. This step is made only after the descending branch has been identified; the proof does not assume that every solution of the area equation lies on that branch.

LEAN REFERENCES

```
SameCurvatureArea.lean LeanSuffixAnalytic.typeThreeArea_lt_typeFourArea;
LeanSuffixAnalytic.typeThreeArea_descending_root_lt
```

Set $\Delta(t) = P_3(r(t)) - P_4(t)$. Continuity and local implicit differentiation give

$$r'(t) = \frac{A_4'(t)}{A_3'(r(t))}, \quad \Delta'(t) = (r(t) - t)A_4'(t).$$

Thus Δ increases up to t_* and decreases afterwards. One-sided continuity handles $t = 1$, where the radical derivative is singular. Therefore

$$\Delta(t) \leq \Delta(t_*) = P_3(s_*) - P_4(t_*) < 0, \quad 0 < t \leq 1.$$

A single stationary certificate consequently produces an equal-area, strictly improving type-(iii) competitor for every admissible type-(iv) curvature at the same density.

LEAN REFERENCES

```
EqualAreaEnvelope.lean LeanSuffixAnalytic.equalAreaRoot_continuousOn_of_compact_unique;
                        LeanSuffixAnalytic.equalAreaRoot_hasDerivAt;
                        LeanSuffixAnalytic.equalAreaPerimeterGap_hasDerivAt;
                        LeanSuffixAnalytic.equalAreaPerimeterGap_le_fold;
                        LeanSuffixAnalytic.allCurvature_typeThreeImprovement_of_envelope
CMVSuffixModel.lean CMVSuffixModel.FourArcCandidate.not_isWeightedPerimeterMinimizer_of_stationaryEqualAreaPair
```

4. Exact certification of the density range

It remains to verify the six stationary conditions uniformly in λ . The proof uses finite exact-rational interval certificates, not floating-point samples. Every interval operation carries a real-containment theorem; sine and cosine are bounded by Taylor polynomials with proved remainders; inverse-trigonometric bounds include principal-branch checks; and centered mean-value enclosures retain shared intermediate points instead of treating repeated occurrences independently.

LEAN REFERENCES

```
LeanSuffixReflective.lean LeanSuffixReflective.QInterval.RealContains
ScalarSuffixCertificate.lean ScalarSuffixCertificate.CertificateExpr.sound;
                             ScalarSuffixCertificate.sinTaylor27Interval_sound;
                             ScalarSuffixCertificate.cosTaylor26Interval_sound;
                             ScalarSuffixCertificate.realContains_arcsin_of_sin_bounds
LeanSuffixAnalytic.lean LeanSuffixAnalytic.centered_meanValue_exists
```

On the lower range, the certificate uses four angular coordinates $x, y, w, v \in (0, \pi/4)$. The identities

$$t_* = \cos x, \quad s_* = \frac{1 + \cos w}{2}$$

encode the two stationary parameters. Two residual equations impose the Snell relations, and two more impose stationarity and equal area. A rational affine predictor and a closed four-dimensional error box reduce existence to a common-zero argument: each residual has opposite signs on the corresponding pair of faces. The formal triangular elimination then proves the existence of the stationary pair, rather than merely locating a numerical approximation.

LEAN REFERENCES

```
ScalarSuffixCertificate.lean ScalarSuffixCertificate.oppositeFace_zero
MiddleFaceAssembly.lean MiddleFaceAssembly.Brick; MiddleFaceAssembly.Model.rootMap;
                         MiddleFaceAssembly.Model.rootMap_continuous; MiddleFaceAssembly.CellCertificate;
                         MiddleFaceAssembly.CellCertificate.sound; MiddleFaceAssembly.Conditions.root_exists;
                         MiddleFaceAssembly.pointEquationSemantics;
                         MiddleFaceAssembly.Conditions.stationaryEqualAreaPair_exists
```

The first prototype covers $[51/50, 102001/100000]$; 113 middle cells cover $[102001/100000, 25781/25000]$; and the right prototype covers $[25781/25000, 33/32]$. Their endpoints agree exactly. Together they exclude every candidate for

$$\frac{51}{50} \leq \lambda \leq \frac{33}{32}.$$

LEAN REFERENCES

```
ScalarSuffixCertificate.lean ScalarSuffixCertificate.FirstPrototypeCell.triangular_root;
                             ScalarSuffixCertificate.FirstPrototypeCell.candidate_not_isWeightedPerimeterMinimizer
MiddleFaceCells.lean MiddleFaceCells.candidate_not_isWeightedPerimeterMinimizer;
                     MiddleFaceCells.candidate_not_isWeightedPerimeterMinimizer_from_51_50
RightPrototypeCell.lean RightPrototypeCell.candidate_not_isWeightedPerimeterMinimizer
```

For $33/32 \leq \lambda \leq 9/7$, 1024 compact boxes enclose s_* and t_* directly. Opposite signs of K_4 locate its unique zero; negativity of K_3 fixes the descending branch; opposite signs of $A_3 - A_4(t_*)$ locate the equal-area root; and a strictly negative interval enclosure of $P_3 - P_4$ completes the comparison.

LEAN REFERENCES

```
CompactCellAssembly.lean CompactCellAssembly.Conditions;
                          CompactCellAssembly.Conditions.typeFourFold_existsUnique_in_slab;
                          CompactCellAssembly.Conditions.typeThreeEqualArea_existsUnique_in_slab;
                          CompactCellAssembly.Conditions.stationaryEqualArea_improvement_exists;
                          CompactCellAssembly.Conditions.candidate_not_isWeightedPerimeterMinimizer
```

The first two boxes are explicit modules. The remaining boxes form a generated family, and the final coverage theorem includes both the explicit and generated cases. Thus no initial interval is omitted.

LEAN REFERENCES

```
CompactFirstCell.lean CompactFirstCell.compactCandidate_not_isWeightedPerimeterMinimizer
CompactSecondCell.lean CompactSecondCell.firstTwoBoxes_candidate_not_isWeightedPerimeterMinimizer
CompactCells1003To1023.lean CompactCells1003To1023.bboxes0To1023_candidate_not_isWeightedPerimeterMinimizer
```

5. High density and the final splice

For a circular segment of unit chord and half-angle $\theta \in (0, \pi)$, define

$$a(\theta) = \frac{\theta - \sin\theta\cos\theta}{4\sin^2\theta}, \quad \ell(\theta) = \frac{\theta}{\sin\theta}.$$

The function a increases continuously from 0 to $+\infty$. Let $\theta(X)$ denote its inverse, set $L(X) = \ell(\theta(X))$, and define

$$g(X) = 2L(X) - L(2X), \quad X > 0.$$

LEAN REFERENCES

```
MorgansArcFunction.lean area; ell; area_strictMonoOn; area_surjOn; 00f; arc; hasDerivAt_arc
GFunction.lean g
```

The function g has a unique global minimum at $X_* = a(\theta_*)$, where $\theta_* \in (0, \pi/2)$ solves

$$3\theta_* - 3\sin\theta_*\cos\theta_* = \pi.$$

An exact rational trigonometric certificate proves

$$\lambda \geq 1.2581840884, \quad X > 0 \quad \Rightarrow \quad \frac{1}{\lambda} < g(X).$$

The decimal is an exact terminating rational, not a measured approximation.

LEAN REFERENCES

```
Minimiser.lean F; 0star; g_0star
MinimumValue.lean g_xstar_le; min_g_eq_three_mul_cos_0star
Certificate.lean cmv_comparison_bound_optimized
```

Let X be the area of one exterior cap divided by c^2 . Replace the two congruent exterior caps by a single upper cap with the same chord and twice the area, retain the strip core, and expose the lower chord. Weighted area is unchanged. The weighted-perimeter improvement is

$$P_\lambda(C) - P_\lambda(C_{\text{rep}}) = \lambda c \left(g(X) - \frac{1}{\lambda} \right) > 0.$$

The lower chord is proved to lie in the new frontier and contributes length c at density 1. Thus the cap replacement is a genuine admissible competitor, not merely a scalar comparison.

LEAN REFERENCES

```
CapReplacement.lean  FourArcCandidate.CapReplacement; FourArcCandidate.capReplacementRegion
CMVGeometry.lean    ReplacementAssembly.exposedLowerChord_subset_frontier
FrontierPerimeter.lean  replacement_frontier_weightedPerimeter_eq
GeometricBridge.lean   FourArcCandidate.cap_replacement_preserves_area;
                       FourArcCandidate.cap_replacement_perimeter_difference;
                       FourArcCandidate.cap_replacement_strictly_improves
CandidateExclusion.lean  cmv_type_four_not_minimizing
```

For $\lambda \geq 51/50$, the proof now splits into three exact ranges: the lower certificates apply through $33/32$, the compact certificates apply through $9/7$, and the cap replacement applies above $9/7$. Every case constructs an admissible carrier with equal weighted area and strictly smaller weighted perimeter. This proves the theorem.

LEAN REFERENCES

```
CMVPublishedCutoff51_50.lean  CMVPublishedCutoff51_50.candidate_not_isWeightedPerimeterMinimizer_from_51_50
```

6. Scope of the result

The value $51/50$ is the lower endpoint of the certificate argument, not a claimed geometric phase transition. The first certified brick begins there, the adjoining families cover the interval up to the cap-replacement regime, and no certificate in this package establishes the required stationary comparison below it. The present theorem therefore gives no conclusion for $1 < \lambda < 51/50$.

A proof of the full conjectural classification would require additional geometric and measure-theoretic work. In particular, one must connect the source paper’s regularity and symmetry reductions to the explicit coordinate constructors, treat interface contact and refraction as well as limiting or degenerate configurations, and relate the complete-frontier perimeter used here to the reduced-boundary perimeter of finite-perimeter sets. The Lean theorem resolves the modeled comparison stated above; it does not replace those source-to-model arguments.

References

[1] A. Cañete, M. Miranda Jr., and D. Vittone, “Some Isoperimetric Problems in Planes with Density,” *Journal of Geometric Analysis* **20** (2010), 243–290; arXiv:0906.1256v1. For the strip problem, see §3.2, especially Lemma 3.8, equations (26)–(27), Conjecture 3.12, Theorem 3.16, and Remark 3.17.